

BrokerVerse OOTB

Insurance broking platform by iorta TechNXT

Solution Architecture · Document 03 of 11

Database Inventory

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Prepared by iorta TechNXT

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Version	Date	Author	Change
1.0	29 September 2026	iorta TechNXT	Initial issue
1.1	29 September 2026	iorta TechNXT	Updated to the current system: placement journey, co-insurance, posting rules, period end and BIR tax, bank reconciliation, document numbering, commission rate matrix, single PDF engine, uploads and go-live imports, housekeeping, business time zone, broker roles, deployment on the BrokerVerse URL

Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document is the inventory of the BrokerVerse OOTB database: every table with its domain, purpose, key columns, row count and size, number of indexes, retention class and owning backend module, followed by the indexes, views, sequences, extensions, functions, triggers, the application settings by group and the scheduled jobs. It supports capacity planning (document 06), backup and archival (documents 09 and 10) and data-protection reviews. The design rationale is in document 02.

1.2 Source and currency of the figures

The inventory is generated from the catalogue of a PostgreSQL 16 database created for this issue (brokerverse_arch, migrated to 0152_email_sender.sql and seeded with SEED_SAMPLE_DATA=true) by docs/architecture/tools/collect_db_inventory.py, which runs read-only catalogue queries and one count(*) per table. The descriptions, owning modules and retention classes come from tools/table_catalog.py.

Note: Row counts and sizes are those of a fresh database with reference data and the sample data set. They show relative weight, not production volumes. A production database starts with reference data only. Re-run the collector against UAT or production and rebuild the documents to refresh them (see docs/architecture/README.md).

Size is the total relation size in KB (table, TOAST and indexes). PostgreSQL allocates at least one 8 KB page per relation and index, so small tables show 16 to 64 KB. "Idx" is the number of indexes including the primary key.

1.3 Retention classes

Class	Meaning and default handling (periods in document 10)
Financial	Ledger, bills, receipts, payments, vouchers, commissions, remittances, debit notes, bank statements and reconciliations, close runs, BIR certificates, reinsurance settlements. Kept for the statutory record-keeping period; cancelled or reversed, never deleted from the online system while the period runs.
Transaction	Leads, clients, quotations, broker slips, offers, placements, policies, endorsements, claims, renewals, treaties, cessions, documents. Kept for the policy or claim life plus the regulatory period.
Reference	Masters, settings, posting rules, number series, tax codes, product templates, roles, permissions, report catalogue, scheduled jobs. Kept while in use; changes audited.
Log / audit	Audit log, sign-in history, histories and field-level trails, job run history, notice logs. Kept for the audit period, then archived.
Temporary	Tokens, reset codes, e-mail outbox, notifications, work queues, generated report files. Purged by the housekeeping job after the periods set in System Settings.

2 Summary

2.1 Tables by domain

Domain	Tables	Rows (test data)	Size KB (test data)	Tables
Security and identity	9	240	312	login_history, password_history, password_resets, permissions, refresh_tokens, role_permissions, roles, user_roles, users
Platform services	14	872	1,120	agent_events, app_settings, audit_log, document_numbering, documents, email_outbox, job_queue, job_runs, master_records, master_types, notifications, scheduled_jobs, schema_migrations, sequences
Reference masters	16	234	704	banks, branches, cities, countries, coverages, currencies, districts, insurance_companies, policy_types, postal_codes, products, signatories, states, vehicle_brands, vehicle_models, vehicle_variants
Product configurator	4	73	280	product_components, product_risk_mappings, product_risk_sections, product_templates
Sales and policy	6	81	872	clients, endorsements, leads, policies, policy_payments, quotes
Placement and co-insurance	4	52	528	broker_slips, insurer_offers, placements, risk_participants
Claims	4	111	248	claim_field_changes, claim_history, claim_settlement_movements, claims
Renewals	7	54	448	renewal_activities, renewal_batch_policies, renewal_batches, renewal_notices, renewal_quotes, renewals, winback_campaigns
Billing, receipts and collections	9	134	696	collection_actions, collection_items, receipt_applications, receipt_lines, receipts, receivable_credits, receivable_participants, receivables, write_off_reasons
General ledger and posting rules	7	556	744	accounting_periods, entry_matches, gl_accounts, journal_lines, journal_vouchers, posting_rule_lines, posting_rules
Period end and tax	12	37	392	bir_2307_certificates, fiscal_years, opening_balances, period_close_checklist, period_close_entries, period_close_run_checks, period_close_runs, period_status_history, recurring_journal_runs, recurring_journals, tax_codes, year_end_runs
Bank reconciliation	9	27	384	bank_match_rules, bank_rec_match_items, bank_rec_matches, bank_reconciliation_history, bank_reconciliations, bank_statement_formats, bank_statement_lines, bank_statements, bank_transaction_types
Disbursements and petty cash	9	32	768	checkbooks, disbursements, invoice_lists, petty_cash_disbursements, petty_cash_funds, petty_cash_receipts, petty_cash_replenishments, petty_cash_request_lines, petty_cash_requests
Commission	4	30	288	commission_adjustments, commission_rates, commission_referrers, commissions
Remittance and direct bill	11	89	696	commission_debit_note_collections, commission_debit_note_lines, commission_debit_notes, direct_bill_items, insurer_refund_credits, remittance_allocations, remittance_approvals, remittance_delegations, remittance_items, remittance_lines, remittances
Reinsurance	7	37	480	cessions, reinsurance_bordereaux, reinsurance_exceptions, reinsurance_reconciliations, reinsurance_recoveries, reinsurance_treaties, reinsurers
Incentive	3	26	160	incentive_calculations, incentive_programs, incentive_results
Reporting	3	39	192	generated_reports, report_definitions, report_schedules
Total	138	2,724	9,312	

2.2 Tables with the most rows (test data)

Table	Rows (test data)	Total size KB	Average row bytes	Retention class
app_settings	351	144	199	Reference
master_records	319	440	394	Reference
journal_lines	210	152	185	Financial
role_permissions	163	24	32	Reference
gl_accounts	129	88	173	Reference
posting_rule_lines	100	80	109	Reference
journal_vouchers	86	264	334	Financial
schema_migrations	58	32	61	Reference
permissions	56	48	78	Reference
master_types	53	104	915	Reference
document_numbering	52	96	182	Reference
cities	49	64	135	Reference
claim_field_changes	48	48	122	Log / audit
claim_history	48	48	88	Log / audit
product_components	48	104	406	Reference

3 Table inventory

Key columns: PK primary key, UK unique constraint, FK foreign-key columns (first four), JSONB document columns. Rows and KB are test data.

Security and identity (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
login_history	Every sign-in attempt: user, IP, user agent, success, reason, method	PK id FK user_id	0	32	3	Log / audit	auth
password_history	Previous password hashes for the history rule (last N kept)	PK id FK user_id	0	24	2	Log / audit	auth
password_resets	One-time password reset codes (HMAC hash only), expiry and attempts	PK id FK user_id	0	24	2	Temporary	auth
permissions	Permission codes read:<module>, write:<module> (27 modules) and approve:<area> (period end, bank reconciliation)	PK id UK code	56	48	2	Reference	users
refresh_tokens	Refresh-token register (jti, family, expiry, rotation / revocation reason)	PK jti FK user_id	0	32	3	Temporary	auth
role_permissions	Grant of permissions to roles	PK role_id+permission_id	163	24	1	Reference	users
roles	Role catalogue: system-admin, sales, processing, operations, claims, accounting, accounting-manager (inherits accounting) and custom roles	PK id UK code	7	48	2	Reference	users
user_roles	Assignment of roles to users	PK user_id+role_id	7	32	1	Reference	users
users	User accounts: bcrypt password hash, status, lockout counter, 2FA (encrypted TOTP secret), token version	PK id UK username	7	48	2	Reference	users, auth

Platform services (14 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
agent_events	Calendar events and reminders of a user (open items screen)	PK id FK user_id	0	24	2	Transaction	payments (open items)
app_settings	Configuration keys (JSONB value, group, label, type); every business parameter	PK key JSONB value	351	144	1	Reference	settings, system-settings
audit_log	Who did what to which record, with before / after JSON and IP	PK id JSONB before_data, after_data	0	40	4	Log / audit	all (lib/audit.js)
document_numbering	Document number series: prefix, pattern, width, reset rule, start number	PK id UK code	52	96	3	Reference	document-numbering
documents	Registry of stored files: storage key, name, type, size, owning entity, uploader	PK id UK storage_key	0	24	2	Transaction	uploads and file writers
email_outbox	Queued e-mails (to, subject, HTML body, status, attempts, error)	PK id	0	24	2	Temporary	lib/mailer.js
job_queue	Background work queue (renewal notice batches); status, progress, attempts	PK id JSONB payload, progress, result	0	24	2	Temporary	renewals
job_runs	Run history of scheduled and manual job runs (status, output, error)	PK id FK job_id JSONB output	0	24	2	Log / audit	jobs
master_records	Generic master records (JSONB data) for masters without their own table	PK id FK type_code JSONB data	319	440	4	Reference	masters

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
master_types	Master screen registry: fields, storage table, keys of each configurable master	PK code JSONB fields, unique_keys	53	104	1	Reference	masters
notifications	In-app notifications per user or per permission audience	PK id FK user_id	0	24	2	Temporary	notifications
scheduled_jobs	Cron jobs (code, cron, handler, params, enabled, last status)	PK id UK code JSONB params	15	48	2	Reference	schedules, jobs
schema_migrations	Applied migration files	PK name	58	32	1	Reference	db/migrate.js
sequences	Document-number counters per series and reset period (year, fiscal year, month or ALL)	PK name+period	24	72	1	Reference	document-numbering

Reference masters (16 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
banks	Banks and mortgagees	PK id UK code JSONB attrs	6	48	2	Reference	masters
branches	Broker branches	PK id UK code JSONB attrs	3	48	2	Reference	masters
cities	Cities and municipalities	PK id FK state_id JSONB attrs	49	64	1	Reference	masters
countries	Countries	PK id UK code JSONB attrs	5	48	2	Reference	masters
coverages	BI / PD / PA cover options with amount and premium	PK id FK policy_type_id JSONB attrs	11	32	1	Reference	quotations
currencies	Currencies (base flag, decimals, exchange rate)	PK id UK code JSONB attrs	5	48	2	Reference	masters
districts	Districts / barangays with postal code	PK id UK city_id+name FK city_id	26	48	2	Reference	addresses
insurance_companies	Insurers (principals): contact, default commission rate, credit terms, billing mode	PK id UK code JSONB attrs	13	48	2	Reference	masters
policy_types	Policy types per product	PK id FK product_id JSONB attrs	14	32	1	Reference	masters
postal_codes	Postal code lookup (province, city, district)	PK id UK country_code+code+district	26	64	3	Reference	addresses
products	Insurance products / lines	PK id UK code JSONB attrs	9	48	2	Reference	masters
signatories	Authorised signatories for documents	PK id JSONB attrs	3	32	1	Reference	masters
states	Provinces / states	PK id FK country_id JSONB attrs	17	32	1	Reference	masters
vehicle_brands	Vehicle makes	PK id UK name JSONB attrs	8	48	2	Reference	quotations
vehicle_models	Vehicle models per make	PK id FK brand_id JSONB attrs	19	32	1	Reference	quotations

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
vehicle_variants	Vehicle variants (body type, seating)	PK id FK model_id JSONB attrs	20	32	1	Reference	quotations

Product configurator (4 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
product_components	Template components: covers, rating tables, rules, documents (JSONB data)	PK id FK template_id JSONB data	48	104	3	Reference	product-configurator
product_risk_mappings	Product / line-of-business risk definitions	PK id UK product_code JSONB configuration	8	48	2	Reference	product-configurator
product_risk_sections	Risk sections of a mapping with default rates	PK id UK mapping_id+section_code FK mapping_id	4	48	2	Reference	product-configurator
product_templates	Versioned product templates (config JSONB, rates, limits, insurers)	PK id FK parent_id, product_id JSONB config, features, insurers	13	80	2	Reference	product-configurator

Sales and policy (6 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
clients	Clients (insured parties) with KYC fields and owner	PK id UK client_code FK lead_id, owner_user_id JSONB extra	20	80	4	Transaction	clients
endorsements	Policy endorsements and cancellations, premium delta, billing link	PK id UK endorsement_number FK client_id, policy_id, receivable_id JSONB changes, endorsement_type_ids, completion	3	96	5	Transaction	endorsements
leads	Leads / prospects (individual or corporate), owner, status	PK id UK lead_number FK client_id, owner_user_id, product_id JSONB extra	16	128	7	Transaction	leads
policies	Issued policies: term, premium, insurer, billing mode, details / doc JSONB	PK id UK policy_number FK client_id, insurance_company_id, lead_id, owner_user_id +5 JSONB details, doc	28	256	13	Transaction	policies
policy_payments	Premium payments captured on a policy, awaiting finance confirmation	PK id FK confirmed_by, policy_id, receipt_id, receivable_id +2	0	48	5	Financial	policies
quotes	Quotations: vehicle / coverage JSONB, premium, taxes, commission, approval token hash	PK id UK quote_number FK agent_user_id, broker_slip_id, client_id, insurance_company_id +5 JSONB vehicle, coverage, doc	14	264	13	Transaction	quotations

Placement and co-insurance (4 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
broker_slips	Broker slips (BS-): a client risk sent to several insurers for terms	PK id UK slip_number FK client_id, lead_id, owner_user_id, product_id +1 JSONB risk_details, doc, requested_covers	2	144	8	Transaction	placement
insurer_offers	Insurer offers (OFR-) on a broker slip: premium, rate, taxes, deductibles, validity, share, or a decline	PK id UK offer_number, broker_slip_id+insurance_com pany_id FK broker_slip_id, insurance_company_id	5	80	4	Transaction	placement
placements	Placement slips (PS-): firm order to the chosen insurers, binding per participant, policy issue	PK id UK placement_number FK broker_slip_id, client_id, insurance_company_id, lead_id +5 JSONB doc	1	208	12	Transaction	placement
risk_participants	Co-insurance participants of a broker slip, quotation, placement or policy: lead, share, premium and commission	PK id UK entity_type+entity_id+insuranc e_company_id FK insurance_company_id	44	96	5	Transaction	placement, policies

Claims (4 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
claim_field_changes	Field-level audit trail of claim edits	PK id FK claim_id	48	48	2	Log / audit	claims
claim_history	Claim status history	PK id FK claim_id	48	48	2	Log / audit	claims
claim_settlement_m ovements	Claim settlement cash: funds received from each insurer and payments to the claimant, with journal	PK id FK claim_id, insurance_company_id, journal_id	0	24	2	Financial	claims
claims	Claims: loss details, estimate, approved and settled amounts, handler, lifecycle status	PK id UK claim_number FK client_id, handler_user_id, policy_id JSONB details, driver, third_party	15	128	7	Transaction	claims

Renewals (7 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
renewal_activities	Contact and follow-up activities on a renewal	PK id FK renewal_id JSONB details	9	48	2	Log / audit	renewals
renewal_batch_polic ies	Policies in a renewal batch and their notice status	PK id UK batch_id+policy_id FK batch_id, policy_id, renewal_id	7	80	4	Transaction	renewals
renewal_batches	Batch renewal notice runs	PK id UK batch_number JSONB criteria	2	48	2	Transaction	renewals
renewal_notices	Renewal notices sent (stage, method, recipient, status)	PK id UK renewal_id+stage FK renewal_id	19	48	2	Log / audit	renewals

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
renewal_quotes	Renewal quotations (loading, discount, variance)	PK id UK quote_number FK renewal_id JSONB taxes, rating	2	64	3	Transaction	renewals
renewals	Renewal cases of expiring policies (status, notices, contact, premium old / new)	PK id FK client_id, new_policy_id, policy_id JSONB coverage_details, accessories, order_summary	14	112	6	Transaction	renewals
winback_campaigns	Win-back campaigns for lapsed business	PK id UK campaign_number JSONB offers	1	48	2	Transaction	renewals

Billing, receipts and collections (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
collection_actions	Collection actions: calls, e-mails, commitments	PK id FK collection_id	5	48	2	Log / audit	collections
collection_items	Collection follow-up items for open bills (commitment, escalation, assignee)	PK id UK receivable_id FK assigned_to, client_id, policy_id	28	96	5	Transaction	collections
receipt_applications	Application of receipt lines to bills, with journal link	PK id FK journal_id, receipt_id, receipt_line_id, receivable_id	23	96	5	Financial	receipts
receipt_lines	Receipt lines per policy (premium, taxes, EWT, discounts)	PK id UK receipt_id+line_no FK policy_id, receipt_id	23	64	3	Financial	receipts
receipts	Official receipts (OR-) header: payer, amount, mode, status	PK id UK receipt_number FK bank_id, client_id, policy_id, receivable_id	23	144	8	Financial	receipts
receivable_credits	Return premium and cancellation credits applied to a bill, with the refund part and journal	PK id FK journal_id, policy_id, receivable_id	0	40	4	Financial	receipts
receivable_participants	Per-insurer split of a co-insured bill: share, gross, commission, taxes, due to insurer	PK receivable_id+insurance_company_id	0	24	2	Financial	receipts
receivables	Premium bills (INV-) per policy / endorsement / renewal with balance and ageing	PK id UK bill_number FK booking_jv_id, client_id, policy_id	28	136	6	Financial	receipts
write_off_reasons	Write-off reasons with GL account and maximum amount	PK id UK code JSONB attrs	4	48	2	Reference	masters, accounting

General ledger and posting rules (7 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
accounting_periods	Periods 1 to 12 and adjustment period 13 per fiscal year: dates and status (open, soft_closed, closed, locked)	PK period FK fiscal_year	0	24	2	Financial	accounting, period-end
entry_matches	Open-item matching of debit and credit journal lines	PK id FK credit_line_id, debit_line_id, write_off_jv_id	0	40	4	Financial	accounting
gl_accounts	Chart of accounts (type, statement group, normal balance, open-item flag)	PK code FK parent_code	129	88	2	Reference	accounting

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
journal_lines	Journal lines: account, debit / credit, branch, department, client / policy tags	PK id FK client_id, jv_id, policy_id	210	152	5	Financial	journal-vouchers, accounting
journal_vouchers	Journal vouchers (JV-): source, status, maker-checker, reversal / correction links	PK id UK jv_number FK client_id, correction_of, policy_id, posting_rule_id +2	86	264	10	Financial	journal-vouchers, accounting
posting_rule_lines	Lines of a posting rule: side, account or account role, amount key, split per co-insurer	PK id UK rule_id+line_no FK rule_id	100	80	2	Reference	posting-rules
posting_rules	Posting rule versions per business event (effective date, entry type, narration)	PK id UK event_code+version	31	96	3	Reference	posting-rules

Period end and tax (12 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
bir_2307_certificates	BIR Form 2307 certificates issued and received, per payee and quarter	PK id UK cert_number JSONB lines	0	32	3	Financial	period-end
fiscal_years	Fiscal years (FY2026): dates and status	PK code	0	24	2	Financial	period-end
opening_balances	Opening balances per fiscal year and account: carried forward by the year-end close or loaded at go-live	PK fiscal_year+account_code	0	16	1	Financial	period-end
period_close_checklist	Month-end checklist items (automatic or manual, severity)	PK code	11	32	1	Reference	period-end
period_close_entries	Journals created by a close run and their automatic reversal	PK id FK jv_id, reversal_jv_id, run_id	0	48	5	Financial	period-end
period_close_run_checks	Checklist results of a close run and sign-off of manual items	PK id UK run_id+code FK run_id JSONB detail	0	24	2	Log / audit	period-end
period_close_runs	Month-end close runs (MEC-): steps, preparer, approval	PK id UK run_number FK period JSONB steps	0	32	3	Financial	period-end
period_status_history	Every change of an accounting period status, with source and user	PK id	0	24	2	Log / audit	period-end
recurring_journal_runs	Journals generated from a recurring template per occurrence	PK id FK jv_id, recurring_id	0	32	3	Financial	period-end
recurring_journals	Recurring and accrual journal templates (frequency, lines, auto-post, auto-reverse)	PK id UK code JSONB lines	0	32	3	Reference	period-end
tax_codes	Tax codes: VAT, expanded withholding with ATC, DST, LGT; rate and GL account	PK code	26	48	2	Reference	period-end
year_end_runs	Year-end close runs (YEC-): closing and transfer journals, net income, reversal	PK id UK run_number FK closing_jv_id, fiscal_year, transfer_jv_id JSONB checks	0	48	5	Financial	period-end

Bank reconciliation (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
bank_match_rules	Automatic matching rules (reference, amount-date, one-to-many, many-to-one)	PK code JSONB params	6	32	1	Reference	bank-reconciliation

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
bank_rec_match_items	Bank and book lines of a match	PK id FK bank_line_id, journal_line_id, match_id	0	40	4	Financial	bank-reconciliation
bank_rec_matches	Matches between bank lines and cash-account journal lines, with difference treatment	PK id	0	24	2	Financial	bank-reconciliation
bank_reconciliation_history	Status history of a reconciliation	PK id FK rec_id	0	24	2	Log / audit	bank-reconciliation
bank_reconciliations	Monthly bank reconciliation statements (BRC-): figures, preparer, approver	PK id UK rec_number JSONB snapshot	0	32	3	Financial	bank-reconciliation
bank_statement_formats	Bank statement file layouts (columns, date format, sign convention)	PK code JSONB columns	4	32	1	Reference	bank-reconciliation
bank_statement_lines	Statement lines (amount, reference, line hash), status and adjustment journal	PK id UK txn_number FK statement_id	9	96	5	Financial	bank-reconciliation
bank_statements	Imported or keyed bank statements per bank account and period, with file hash	PK id UK statement_number	1	72	4	Financial	bank-reconciliation
bank_transaction_types	Bank-side transaction types (charges, interest, direct credits) and their GL account role	PK code	7	32	1	Reference	bank-reconciliation

Disbursements and petty cash (9 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
checkbooks	Cheques issued (instrument, approval, printing)	PK id FK disbursement_id, invoice_list_id, journal_id	3	80	4	Financial	disbursements
disbursements	Payment vouchers (PV-) to insurers, agents, clients; approval and payment	PK id UK voucher_number FK bank_id, client_id, insurance_company_id, journal_id +1	16	192	9	Financial	disbursements
invoice_lists	Invoice lists (IL-) settled by a disbursement	PK id UK invoice_number FK client_id, disbursement_id, insurance_company_id, policy_id +1	5	144	8	Financial	disbursements
petty_cash_disbursements	Petty cash payments (VAT, WHT, journal)	PK id UK transaction_number FK fund_id, journal_id, request_id	1	96	5	Financial	payments
petty_cash_funds	Petty cash funds (size, limits, custodian, available cash)	PK id UK code FK custodian_user_id, journal_id	2	48	2	Financial	payments
petty_cash_receipts	Receipts into a petty cash fund	PK id UK receipt_number FK fund_id, journal_id	0	40	4	Financial	payments
petty_cash_replenishments	Fund replenishments from bank	PK id UK transaction_number FK fund_id, journal_id	0	40	4	Financial	payments
petty_cash_request_lines	Lines of a petty cash request	PK id FK request_id	3	48	2	Financial	payments
petty_cash_requests	Petty cash requests and approval	PK id UK request_number FK fund_id, requester_user_id	2	80	4	Financial	payments

Commission (4 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
commission_adjustments	Commission reversed or clawed back when premium is returned or a policy cancelled	PK id FK commission_id, journal_id, policy_id	0	32	3	Financial	commission
commission_rates	Commission rate matrix: rate per insurer, product, line and policy type with effective dates	PK id FK insurance_company_id, product_id	0	24	2	Reference	commission-rates
commission_referrers	Referrers / sub-agents (no sign-in): hierarchy, WHT rate, bank account	PK id FK parent_referrer_id, user_id	8	32	1	Reference	commission
commissions	Commission lines per policy / endorsement and referrer: accrual, eligibility, approval, payment	PK id FK accrual_jv_id, agent_user_id, disbursement_id, endorsement_id +4	22	200	9	Financial	commission, policies

Remittance and direct bill (11 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
commission_debit_note_collections	Insurer payments against a debit note (cash, EWT, Form 2307)	PK id UK collection_number FK debit_note_id, journal_id	0	40	4	Financial	remittance (direct bill)
commission_debit_note_lines	Policies / items on a debit note	PK id FK debit_note_id, item_id, policy_id	0	40	4	Financial	remittance (direct bill)
commission_debit_notes	Commission debit notes (DN-) to insurers for direct-billed policies	PK id UK dn_number FK insurance_company_id	0	32	3	Financial	remittance (direct bill)
direct_bill_items	Commission receivable items booked for direct-billed policies	PK id FK booking_jv_id, debit_note_id, endorsement_id, insurance_company_id +2	0	64	7	Financial	remittance (direct bill)
insurer_refund_credits	Refunds due from insurers on returned premium already remitted; netted against the next remittance	PK id FK insurance_company_id, journal_id, policy_id JSONB split, applications	0	32	3	Financial	remittance
remittance_allocations	Premium collected on a co-insured policy, split for remittance to each insurer	PK id UK receipt_application_id+insurance_company_id FK insurance_company_id, invoice_list_id, receipt_application_id	0	40	4	Financial	receipts, remittance
remittance_approvals	Multi-level approval requests with SLA and history	PK id FK action_by, delegated_to, initiator_id JSONB history	19	112	4	Log / audit	remittance
remittance_delegations	Approval delegations (period, transaction types, amount limit)	PK id FK delegate_id, delegator_id JSONB trans_types	1	32	1	Reference	remittance
remittance_items	Remittance work items (settlements, adjustments, transfers, exceptions)	PK id UK reference_no FK created_by, insurance_company_id, remittance_id JSONB data	18	128	5	Financial	remittance
remittance_lines	Policies included in a remittance	PK id FK insurance_company_id, policy_id, remittance_id	34	80	4	Financial	remittance

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
remittances	Premium remittances / bills to insurers (REM-): gross, commission, net due, approval, settlement	PK id UK remittance_number FK agent_user_id, insurance_company_id JSONB delivery_method, data	17	96	5	Financial	remittance

Reinsurance (7 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
cessions	Cessions of policies to treaties / facultative reinsurers	PK id FK facultative_reinsurer_id, policy_id, treaty_id	13	96	5	Transaction	reinsurance
reinsurance_bordereaux	Premium / claims bordereaux (entries JSONB, file)	PK id UK reference FK reinsurer_id, treaty_id JSONB totals	2	80	4	Transaction	reinsurance
reinsurance_exceptions	Reconciliation exceptions and resolution	PK id FK reconciliation_id	2	48	2	Log / audit	reinsurance
reinsurance_reconciliations	Reinsurer statement reconciliations	PK id UK reference FK reinsurer_id	2	64	3	Financial	reinsurance
reinsurance_recoveries	Claim recoveries from reinsurers	PK id UK recovery_number FK cession_id, claim_id, treaty_id JSONB cash_call, documents	6	96	5	Financial	reinsurance
reinsurance_treaties	Treaties (type, capacity, share, retention, approval)	PK id JSONB reinsurer_ids, terms	6	48	2	Transaction	reinsurance
reinsurers	Reinsurers (rating, capacity, contact)	PK id JSONB contact	6	48	2	Reference	reinsurance

Incentive (3 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
incentive_calculations	Incentive calculation batches with maker-checker and payment	PK batch_id FK approved_by, created_by, rejected_by, submitted_by JSONB programs_included	4	32	1	Financial	incentive
incentive_programs	Incentive programmes (metric, targets, tiers, eligibility)	PK id JSONB applicable_to, structure, eligibility	4	48	2	Reference	incentive
incentive_results	Incentive results and payouts per participant	PK id UK program_id+agent_user_id+period FK agent_user_id, calculation_id, program_id	18	80	4	Financial	incentive

Reporting (3 tables)

Table	Purpose	Key columns	Rows	KB	Idx	Retention	Owner module
generated_reports	Generated report files (code, parameters, format, storage key, rows)	PK id JSONB params, totals	0	32	3	Temporary	reports
report_definitions	Report catalogue (parameters, query, columns, roles, permission)	PK code JSONB parameters, default_columns	39	144	1	Reference	reports
report_schedules	E-mailed report schedules (cron, format, recipients)	PK id FK job_id, report_code JSONB params	0	16	1	Reference	reports

4 Other database objects

4.1 Database facts

Item	Value (local test database)
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
Database size	20.7 MB (test data)
Server time zone	Asia/Manila
Tables (schema public)	138
Columns by type	text 1320, timestamp with time zone 310, numeric 256, integer 139, date 99, jsonb 92, bigint 47, boolean 44, ARRAY 5
Indexes	448 (229 unique incl. 138 primary keys, 33 partial, 1 GIN)
Foreign keys	234 (36 without a supporting index)
Identity sequences (serial / identity columns)	63
Document-number counters (table sequences)	24 counters for 2026
Extensions	pgcrypto 1.3, plpgsql 1.0
Application functions	accounting_period_lock_guard, backfill_risk_participants, document_numbering_mirror_prefix, format_document_number, gl_accounts_defaults, jv_check_posting, next_document_number, next_number, numbering_business_date, numbering_fiscal_year, numbering_optional_token, numbering_period_key, numbering_prefix, pe_balance_before, quotes_link_product, rp_insurer_id, rp_parts_from_doc, rp_share, rp_write_split, rpt_age_bucket, set_updated_at, user_effective_roles
Triggers	accounting_period_lock_guard_trg (accounting_periods), commission_rates_updated (commission_rates), document_numbering_mirror (document_numbering), document_numbering_updated (document_numbering), gl_accounts_defaults_trg (gl_accounts), jv_check_posting_trg (journal_vouchers), quotes_link_product (quotes), roles_updated (roles), users_updated (users)
Views / materialised views	bank_account_links, bank_book_lines
Migrations applied	58 (0001_core.sql to 0152_email_sender.sql)
Application settings (app_settings)	351 keys in 36 groups
Scheduled jobs (scheduled_jobs)	15
Snapshot taken	2026-09-29T16:04:56+00:00

4.2 Indexes

448 indexes: 138 primary keys, 91 further unique indexes, 33 partial indexes and 1 GIN index (`master_records_data_idx` on `master_records.data`). The partial indexes serve hot subsets and conditional uniqueness: open direct-bill items, live refresh tokens, active entry matches, direct-billed policies, due recurring journals and close entries, active bank reconciliations and close runs, one lead co-insurer per record, one open renewal per policy, one pending approval per entity, one active document-number prefix, and codes that are unique among records not deleted.

Foreign keys without a supporting index

Migration `0146_foreign_key_indexes.sql` indexed the foreign keys of every table that grows. The 36 foreign keys below have no index whose first column is the foreign-key column. They are on small master

and set-up tables or are maker-checker stamp columns, and are left without an index on purpose. `node backend/scripts/fk-index-report.js` lists them again after new migrations.

Table	Foreign-key columns without an index whose first column is the FK column
cities	state_id
claim_settlement_movements	insurance_company_id, journal_id
commission_adjustments	journal_id
commission_rates	product_id
commission_referrers	parent_referrer_id, user_id
coverages	policy_type_id
gl_accounts	parent_code
incentive_calculations	approved_by, created_by, rejected_by, submitted_by
insurer_refund_credits	journal_id
opening_balances	account_code
petty_cash_funds	custodian_user_id, journal_id
policy_payments	confirmed_by, rejected_by, submitted_by
policy_types	product_id
product_components	template_id
product_templates	parent_id, product_id
remittance_approvals	action_by, delegated_to
remittance_delegations	delegate_id, delegator_id
remittance_items	created_by
report_schedules	job_id, report_code
role_permissions	permission_id
states	country_id
user_roles	role_id
vehicle_models	brand_id
vehicle_variants	model_id

Log tables and their indexes

Table	Indexes	Use
audit_log	PK; (entity, entity_id); (at); (username)	The audit trail endpoint (GET <code>/api/settings/audit</code>) filters by entity, entity id, username and a from / to business-date range, newest first, at most 1,000 rows.
login_history	PK; (user_id, at DESC); (at DESC)	Sign-in reviews and the housekeeping purge by date.
notifications	PK; (user_id, is_read, created_at DESC)	The per-user bell and unread count. Read notifications are purged by housekeeping.
email_outbox	PK; (status, created_at)	The outbox job selects <code>status = 'queued'</code> every 5 minutes; housekeeping purges sent and failed messages by date.
job_runs	PK; (job_id, started_at)	The duplicate-slot check of every scheduled run, the run history screen and the housekeeping purge.
refresh_tokens	PK; live tokens per user (partial); family	Expired and revoked rows are purged by housekeeping after 30 days.
generated_reports	PK; (created_at); (code)	Purged after <code>reports.retention_days</code> by the daily-reports job.

4.3 Views

Two views support bank reconciliation: `bank_account_links` (bank account masters with their GL cash account) and `bank_book_lines` (posted journal lines on bank cash accounts, the book side of matching). There are no materialised views.

4.4 Sequences

63 identity sequences (`<table>_id_seq`) back the serial primary keys. Document numbers do not use PostgreSQL sequences but the counter table `sequences`, one row per series and reset period (document 02).

4.5 Extensions, functions and triggers

Object	Detail
Extensions	plpgsql 1.0 (built in), pgcrypto 1.3 (<code>gen_random_bytes()</code> for primary-key defaults)
<code>next_document_number(code, branch, lob, date)</code> , <code>format_document_number()</code> , <code>numbering_*</code> helpers	Allocate and format the next number of a document series; <code>numbering_business_date()</code> gives today's date in <code>general.timezone</code> and is the default of date columns
<code>next_number()</code> , <code>numbering_prefix()</code>	Older numbering helpers kept for seed scripts
<code>document_numbering_mirror_prefix()</code> / trigger <code>document_numbering_mirror</code>	Copies a series prefix into the read-only setting <code>numbering.<code>.prefix</code>
<code>jv_check_posting()</code> / trigger <code>jv_check_posting_trg</code> on <code>journal_vouchers</code>	Refuses to post an unbalanced journal or one in a period that does not accept postings
<code>accounting_period_lock_guard()</code> / trigger on <code>accounting_periods</code>	Refuses to reopen a period locked by the year-end close
<code>pe_balance_before()</code>	Account balance before a date, including the opening balances of the fiscal year
<code>user_effective_roles()</code>	A user's roles plus the roles they inherit (<code>roles.inherits</code>)
<code>rp_*</code> functions, <code>backfill_risk_participants()</code>	Co-insurance share and split helpers used by migrations and reports
<code>quotes_link_product()</code> / trigger <code>quotes_link_product</code>	Links a quotation to its product
<code>rpt_age_bucket(age, buckets)</code>	Ageing bucket label for reports
<code>gl_accounts_defaults()</code> / trigger <code>gl_accounts_defaults_trg</code>	Derives the normal balance of an account from its type
<code>set_updated_at()</code> / triggers on users, roles, commission_rates, document_numbering	Maintain <code>updated_at</code>

4.6 Application settings by group

`app_settings` holds 351 configuration keys in 36 groups, seeded from `backend/src/db/seeds/settings.json`, SQL seed files and migrations, and edited in Master > Configuration and System Settings. Every key the code reads is seeded (`npm run check:settings`, `test/configuration.test.js`).

Group	Keys	Group	Keys	Group	Keys
accounting	52	endorsements	1	premium	3
bank_reconciliation	6	finance	5	product	4
bir	6	general	11	quotations	7
branding	6	housekeeping	9	quote	4
claims	16	incentive	5	reinsurance	6
collections	7	leads	5	remittance	13
commission	8	limits	7	renewals	23
currency	4	notification	5	reports	19
dashboard	3	numbering	52	security	18
direct_bill	10	placement	8	system	1
documents	4	policies	3	tax	5
email	7	policy	4	uploads	4
Total	351				

Settings relevant to operations: `security.*` (password policy, rate limits, two-factor roles, scoped roles), `limits.*` (lockout, idle time-out, quote validity, renewal notice days), `notification.*` (e-mail on or off, sender), `housekeeping.*` (retention days per table), `reports.retention_days` (90), `reports.download_link_ttl_hours` (72), `reports.max_rows` (50,000), `uploads.*` (allowed types and sizes), `general.timezone` (Asia/Manila), `accounting.account.*` (Account Determination), `bank_reconciliation.*`, `bir.*`, `placement.*` and `numbering.*` (prefix mirrors of the 52 document series).

4.7 Scheduled jobs

15 rows in `scheduled_jobs`, 10 of them enabled (document 01, section on scheduled jobs). Report schedules add rows with the code `report-<id>`.

5 Related documents

This document is one of eleven that together form the BrokerVerse OOTB Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design	02_BrokerVerse_Database_Design.pdf
03	Database Inventory (this document)	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

6 Glossary

Terms and abbreviations used in this document.

Term	Meaning
Account determination	Settings accounting.account.<role> that give the GL account of each account role (cash in bank, premium receivable, due to insurer, commission income ...) used by the posting rules.
ATC	Alphanumeric tax code of the BIR for a type of withholding, printed on BIR Form 2307.
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
bcrypt	Adaptive password-hashing function (bcryptjs library) used for user passwords.
BIR	Bureau of Internal Revenue (Philippines): tax authority whose rules drive record keeping and withholding tax certificates.
BIR Form 2307	Certificate of creditable tax withheld at source, issued by the payor to the payee each quarter.
Broker slip	Request for quotation (BS-) that the broker sends to several insurers for one client risk; the answers are recorded as insurer offers.
Cession	The share of a policy's risk and premium passed to a reinsurer under a treaty or facultatively.
Co-insurance	Placement of one risk with several insurers, one of them the lead, each taking a share; premium, commission and claims are split by share.
Cron	Time-based schedule expression (minute hour day month weekday) used by the scheduled jobs.
Debit note (DN)	Commission debit note raised by the broker to an insurer for direct-billed policies.
Direct bill	Billing mode in which the insurer collects the premium and the broker bills the insurer for its commission.
DST	Documentary Stamp Tax (Philippines), part of the gross premium.
Endorsement	Change to an issued policy (details, cover, premium or cancellation).
EWT	Expanded Withholding Tax (Philippines), e.g. withheld by an insurer on commission (BIR Form 2307).
Fiscal year	Accounting year (FY2026) with twelve periods and an adjustment period 13 used by the year-end close.
FK	Foreign key: column that references the primary key of another table.
GIN	Generalised inverted index (PostgreSQL), used for JSONB searches.
HMAC	Keyed-hash message authentication code; signs file links and hashes one-time codes.
Housekeeping	Daily job that deletes operational rows (job runs, sent e-mails, sign-in history, expired tokens, read notifications) after the retention days in System Settings.
JSONB	PostgreSQL binary JSON column type, used for flexible document parts of records.
JV	Journal voucher: balanced set of debit and credit ledger lines.
KYC	Know your customer: identification data required before a policy is issued.
LGT	Local Government Tax (Philippines), part of the gross premium.
Maker-checker	Control in which a record created or submitted by one user must be approved by a different user.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
Month-end close	Run (MEC-) that posts the accruals, recurring journals and valuations of a period, checks the checklist and, once approved, closes the period.
OR	Official receipt issued for premium received.
PK	Primary key: column(s) that uniquely identify a row.

Term	Meaning
Posting rule	Versioned definition of the journal lines for one business event (policy issued, receipt applied ...); every system journal is built from it.
PV	Payment voucher (disbursement) to an insurer, referrer, client or supplier.
Receivable / bill	Premium bill (INV-) raised when a policy, endorsement or renewal is issued.
Remittance	Payment of collected premium, net of commission, from the broker to the insurer.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
SLA	Service level agreement / target time, e.g. for remittance approvals.
Token version	users.token_version, carried in every access token; raising it ends all sessions of the user.
TOTP / 2FA	Time-based one-time password (RFC 6238) used for two-factor authentication.
Treaty	Reinsurance contract under which risks are ceded automatically within agreed limits.
UAT	User acceptance testing environment.
VAT	Value-added tax (12 % in the Philippines).
WHT	Withholding tax deducted from payments such as referrer commission.
Year-end close	Run (YEC-) that posts the closing entries in period 13, carries the balances into the next year and locks the year.